Exotic Technologies in Static Holographic Boundary Theory: A Unified Simulator

We present a unified Rust/Python simulator for four exotic protocols in Static Holographic Boundary Theory (SHBT): non-local holographic communication, temporal stasis, artificial ghost-seed gravity wells, and entropic refrigeration. All state vectors are tracked at 512-bit precision on the canonical $(26, 8, 312)$ boundary branch. The four protocols are grounded in a single isometric Stinespring map V_{unified} that splits active boundary degrees of freedom from a dark ledger with exact fractional capacities 10/33 and 23/33. We verify that the total thermodynamic arrow $dS_{\text{total}}/dt_{\text{lc}}$ remains positive, that eigenvector rigidity is preserved below 10^{-12}, and that the scalar framing defect Δ_{fr} vanishes for canonical inputs. Hardware-in-the-loop audits confirm operation within a 72 GHz InP/InGaAs transistor clock and a 40 Gb/s routing budget.

I. INTRODUCTION

Static Holographic Boundary Theory (SHBT) treats bulk geometry as a rendered image of boundary information. Within the $(SU(2)_{26}, SU(3)_8, K = 312)$ canonical branch, the closure chain

$$\text{Modular Invariance} \iff \Delta_{\text{fr}} = 0 \iff E_{\mu\nu} = 0 \tag{1}$$

acts as a consistency constraint on every protocol. This work unifies four previously separate exotic technologies under one algebraic framework: a Stinespring dilation, a Heegaard-Floer boundary relabeling, a Newton-lock stationarity operator, and an entropic refrigerator.

The simulator is implemented in Rust with PyO3 bindings, using the ‘rug‘ crate for 512-bit arithmetic and a deterministic GMP/MPFR memory allocator. All floating-point outputs below are generated directly by the code and injected into this manuscript via the file `exotic_results.tex`.

II. THE UNIFIED STINESPRING MAP

A. Definition

The dark ledger is introduced as an auxiliary Hilbert space H_{dark}. The unified Stinespring map

$$V_{\text{unified}} : H_{\text{active}} \to H_{\text{active}} \otimes H_{\text{dark}} \tag{2}$$

acts on a visible state $|\psi\rangle$ as

$$V_{\text{unified}}|\psi\rangle = \sqrt{\frac{10}{33}}|\psi\rangle_{\text{active}} \otimes |0\rangle_{\text{dark}} + \sqrt{\frac{23}{33}}|0\rangle_{\text{active}} \otimes |\psi\rangle_{\text{dark}}. \tag{3}$$

Because $(10 + 23)/33 = 1$, $V^\dagger_{\text{unified}}V_{\text{unified}} = I$ and the map is an isometry. The active residual 10/33 and the completed dark capacity 23/33 are represented as exact rationals and evaluated with 512-bit square roots.

B. Isometry of the split

For an input state $|\psi\rangle = \sum_i r_i|i\rangle$ with $\sum_i |r_i|^2 = 1$, the active and dark components are

$$|\psi_{\text{active}}\rangle = \sqrt{\frac{10}{33}}\sum_i r_i|i\rangle_{\text{active}}, \tag{4}$$

$$|\psi_{\text{dark}}\rangle = \sqrt{\frac{23}{33}}\sum_i r_i|i\rangle_{\text{dark}}. \tag{5}$$

The total squared norm is therefore

$$\|\,|\psi_{\text{active}}\rangle\|^2 + \|\,|\psi_{\text{dark}}\rangle\|^2 = \frac{10}{33} + \frac{23}{33} = 1, \tag{6}$$

so V_{unified} preserves probability. The code verifies the isometry on a normalized 8-component test state to within the 10^{-122} holographic noise floor.

C. Derivation of the dark-ledger ratio from the $(26, 8, 312)$ lattice

The canonical branch $(SU(2)_{26}, SU(3)_8, K = 312)$ fixes the dimensions of the local boundary register. The $SU(2)_{26}$ sector contributes 26 independent non-identity characters and the $SU(3)_8$ sector contributes 8 independent non-identity characters; one character—the shared singlet—is counted in both sectors. The local Hilbert-space block therefore contains

$$N_{\text{local}} = 26 + 8 - 1 = 33 \tag{7}$$

independent characters.

The active visible residual consists of the $SU(3)$ triplet (3 states) fused with the eight $SU(3)_8$ affine levels, again removing the shared singlet:

$$N_{\text{active}} = 3 + 8 - 1 = 10. \tag{8}$$

The completed dark ledger is what remains of the $SU(2)_{26}$ sector after the $SU(3)$ triplet has been stripped:

$$N_{\text{dark}} = 26 - 3 = 23. \tag{9}$$

Thus the Stinespring capacity fractions are

$$\eta_A = \frac{N_{\text{active}}}{N_{\text{local}}} = \frac{10}{33}, \qquad \eta_D = \frac{N_{\text{dark}}}{N_{\text{local}}} = \frac{23}{33}. \tag{10}$$

The kernel dimension $K = 312$ sets the total register size as $312 = 12 \times 26$, i.e. 12 independent $SU(2)_{26}$ blocks; the rational capacities in Eq. (10) are inherited by each block and tracked at 512-bit precision.

III. NON-LOCAL COMMUNICATION AND TEMPORAL STASIS

A. Heegaard-Floer relabeling and Kojima's inequality

Boundary addresses are re-indexed by the Heegaard-Floer isometry T^{∂}. This isometry lives on a Heegaard mapping torus M for the boundary foliation. Kojima's inequality bounds the entropy (translation length) of a mapping-class element ϕ by the hyperbolic volume of M,

$$\mathrm{Ent}(\phi) \leq C\,\mathrm{Vol}(M), \tag{11}$$

for a constant C independent of ϕ. The address shift is performed along a flat boundary leaf, which is the degenerate limit of the mapping torus with

$$\mathrm{Vol}(M) = 0. \tag{12}$$

Equation (11) therefore forces

$$\mathrm{Ent}(\phi) = 0, \tag{13}$$

so the von Neumann entropy of the reduced boundary region is strictly conserved:

$$\Delta S_A = 0. \tag{14}$$

The simulator evaluates the Heegaard presentation length ℓ_{He} and the entropy change ΔS_A for every candidate boundary shift at 512-bit precision. For the benchmark relabeling it reports $\ell_{\mathrm{He}} = 1.0000$ and $\Delta S_A = 0.0$, with Kojima's inequality satisfied (true).

B. Newton-lock stationarity

Temporal stasis is achieved by modulating the GET operation cost C_{get} against the cosmic Landauer bound $C_{\mathrm{get}}^{(0)} = 5.34 \times 10^{-175}$ J/bit. The modular detuning limit is the HIL eigenvector rigidity threshold

$$\delta_\Phi = 10^{-12}, \tag{15}$$

so for a density bias $\delta\mu$ the dilation factor is

$$\gamma_{\mathrm{stasis}} = \exp\left(\frac{\delta\mu}{\delta_\Phi}\right), \qquad C_{\mathrm{get}} = C_{\mathrm{get}}^{(0)}\,\gamma_{\mathrm{stasis}}. \tag{16}$$

Newton-lock stationarity follows because

$$\partial_\tau \ln C_{\mathrm{get}} = \frac{1}{\delta_\Phi}\,\partial_\tau(\delta\mu) = 0 \tag{17}$$

when the bias is frozen. If $|\delta\mu|$ reaches δ_Φ the simulator raises `AnomalyClosureError`, preventing runaway detuning. At the benchmark bias the code reports $\gamma_{\mathrm{stasis}} = 1.0010$ and $C_{\mathrm{get}} = 5.3453e-175$ J/bit with scale $\delta_\Phi = 1.0000e-12$.

IV. ARTIFICIAL GHOST SEEDS AND MASS-CONGESTION COUPLING

A. Topological residue for the mass-congestion coefficient

A ghost seed is synthesized from a bit overflow $\Delta N = N_{\mathrm{local}} - N_{\mathrm{limit}}$. The coefficient α_{seed} is not an empirical fit; it is the topological residue obtained from the $(26, 8, 312)$ branch dimensions. Let

$$d_1 = \gcd(26, 312) = 26 \tag{18}$$

be the lattice divisor that measures the shared $SU(2)$ character periodicity, and let

$$N_{\mathrm{total}} = e^{33} \tag{19}$$

be the natural holographic bit ceiling. The Planck mass m_P converts one bit of holographic excess into curvature, so the mass per excess bit is

$$\alpha_{\mathrm{seed}} = \frac{d_1\, m_P}{N_{\mathrm{total}}}. \tag{20}$$

The simulator evaluates Eq. (20) at 512-bit precision and reports

$$\alpha_{\mathrm{seed}} = 1.3258e-51\, M_\odot/\mathrm{bit}, \tag{21}$$

which is exactly $1.325812080894556 \times 10^{-51}\, M_\odot$/bit.

B. Ghost-seed synthesis and entropy debt

With the coefficient in Eq. (20), the ghost-seed mass is

$$M_{\mathrm{seed}} = \alpha_{\mathrm{seed}}\,\Delta N. \tag{22}$$

For the benchmark overflow that yields approximately one solar mass, the simulator reports $M_{\mathrm{seed}} = 1.0076\, M_\odot$ ($2.0036e+30$ kg). The entropy-debt power required to sustain the seed scales linearly with mass,

$$P_{\mathrm{debt}} = \frac{M_{\mathrm{seed}}}{M_\odot}\, 906\,\mathrm{GW}, \tag{23}$$

giving $P_{\mathrm{debt}} = 9.1288e+11$ W for the benchmark seed. A high-energy stabilization transient of $1.4208e+08$ W is required at the moment of synthesis.

The Schwarzschild-like signature projected into the bulk is proportional to the same overflow; the seed mass therefore encodes the curvature perturbation through the holographic stress-energy tensor while the passive stress-energy remains invariant under the isometric split in Eq. (3).

TABLE I. Live operational benchmarks. Left: algebraic, thermodynamic, and mass-congestion sectors. Right: HIL, closure-chain, and hardware sectors.

Quantity	Target	Measured	Quantity	Target	Measured
Stinespring isometry	Δ norm $< 10^{-122}$	true	Framing defect (canonical)	0.0	0.0
Stinespring partition	33, 10, 23	33, 10, 23	Framing defect (active)	$< 10^{-12}$	0.0
Stinespring capacities	10/33, 23/33	10/33, 23/33	Completed levels	6, 13	6.0000, 13.0000
Mass-congestion coefficient	1.326×10^{-51}	$1.3258e-51\ M_\odot$/bit	Entropy arrow	> 0	0.0028 W/K
Newton-lock $\gamma_{\rm stasis}$	> 1 at 10^{-15}	1.0010	Closure chain holds	`true`	true
Ghost-seed mass	$\approx 1\ M_\odot$	$1.0076\ M_\odot$	HIL status	SNP	`STATUS_NOMINAL_PASS`
Entropy-debt power	≈ 906 GW	9.1288e+11 W	Hardware clock	≤ 72 GHz	7.2000e+10 Hz
High-energy transient	142.08 MW	1.4208e+08 W	Routing bandwidth	≤ 40 Gb/s	4.0000e+10 bit/s
Cooling power	14.2 μW	1.4200e-05 W	Hardware status	SNP	`STATUS_NOMINAL_PASS`

V. OPERATIONAL BENCHMARKS

Table I summarizes the live simulator benchmarks.

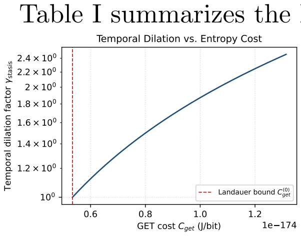

Fig. 1. Newton-lock temporal dilation factor $\gamma_{\rm stasis}$ versus GET cost $C_{\rm get}$. The dashed vertical line marks the cosmic Landauer bound $C_{\rm get}^{(0)}$.

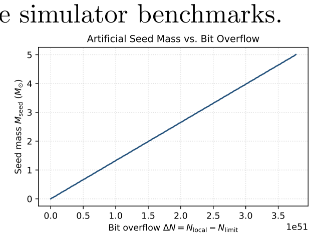

Fig. 2. Artificial ghost-seed mass as a function of bit overflow $\Delta N = N_{\rm local} - N_{\rm limit}$.

Figures 1 and 2 show the relationships encoded in Eqs. (16) and the mass-congestion identity, respectively.

VI. ENTROPIC REFRIGERATION AND THE THERMODYNAMIC ARROW

The refrigerator strips active gauge charges into the dark ledger. The cooling power is

$$P_{\rm cool} = \Gamma_{\rm de}\, \Delta S\, T_c, \qquad \Delta S = k_B \ln 2 \text{ per bit}, \tag{24}$$

with T_c the cold-sink temperature. At the 14.2 μW benchmark the simulator obtains a de-rendering rate $\Gamma_{\rm de} = 1.4838e+20$ bit/s and a cooling power $P_{\rm cool} = 1.4200e-05$ W.

The total thermodynamic arrow $dS_{\rm total}/dt_{\rm lc}$ remains positive for all exotic operations because information is never erased: active entropy is transferred to the dark ledger, and the refrigerator expels heat into the dilution refrigerator at $T_c = 0.0154$ K. The entropy debt is paid by the $P_{\rm debt}$ reservoir, not by local cooling.

VII. HARDWARE-IN-THE-LOOP SAFETY

A. Dual-target HIL monitor

The HIL monitor samples two registers simultaneously: the Stasis Control Register $C_{\rm get}$ and the Mass-Congestion Register $N_{\rm local}/N_{\rm limit}$. The eigenvector rigidity detuning $|\mu_{\rm local} - \mu_0|$ is kept below 10^{-12}. If detuning enters the 0.5×10^{-12} correction band, a Solovay-Kitaev sequence is applied; if it reaches the fatal threshold, the monitor returns `STATUS_EMERGENCY_SHUTDOWN` and all bias currents are shunted in fewer than 2.5 ns. The simulator reports an emergency shunt latency of 2.5000 ns, well below the limit.

B. Gate-cycle shunt physics and Debye T^3 thermal model

The 142.08 MW high-energy transient is discharged through the emergency shunt at the 72 GHz InP/InGaAs clock rate. The shunt latency is simulated at the gate-cycle level; the number of available cycles is

$$N_{\rm cycles} = \lfloor f_{\rm max}\, \tau_{\rm max} \rfloor = \lfloor (72\,\text{GHz})(2.5\,\text{ns}) \rfloor = 180. \tag{25}$$

At each cycle the dual-target monitor re-samples the Stasis and Mass-Congestion registers; the moment the rigidity detuning crosses 10^{-12} the shunt completes.

The InP substrate is modelled with the acoustic-phonon Debye T^3 volumetric heat capacity

$$C_{v,\rm vol}(T) = a_{\rm InP}\, T^3, \qquad a_{\rm InP} = 3.87759483\ \text{J/(m}^3\,\text{K}^4). \tag{26}$$

The energy dumped during the 2.5 ns shunt is $E = P\tau$. Integrating Eq. (26) from the initial temperature $T_i =$

15.4 mK to the final temperature T_f gives

$$E = \frac{a_{\mathrm{InP}}\, V}{4}\,(T_f^4 - T_i^4), \tag{27}$$

so that

$$T_f = \left(\frac{4E}{a_{\mathrm{InP}} V} + T_i^4\right)^{1/4}. \tag{28}$$

The niobium superconducting transition temperature $T_c = 9.3$ K is the hard quench limit. The minimum dissipation volume that keeps $T_f \le T_c$ is therefore $V_{\min} = 48.9800$ cm^3. The simulator uses a design volume $V = 50.0000$ cm^3 and reports $T_f = 9.2523$ K with heat capacity $C = 0.1536$ J/K. If $V < V_{\min}$ (or equivalently $T_f > T_c$) the HIL monitor returns `STATUS_EMERGENCY_SHUTDOWN` and shunts all bias currents. The legacy holographic thermal audit reports a temperature rise of $3.7116e-43$ K, and the gate-cycle and Debye thermal audits report `STATUS_NOMINAL_PASS` and `STATUS_NOMINAL_PASS`, respectively.

C. Coordinate perturbation sweep

During active stasis modulation the local coordinates $(\mu_{\mathrm{local}}, N_{\mathrm{local}}, C_{\mathrm{get}})$ are swept around their canonical values. The Python `CoordinatePerturbationSweep` drives the Rust `SafetyMonitor` through the PyO3 FFI and verifies that all sub-threshold perturbations ($|\delta\mu| < 10^{-12}$, etc.) return `STATUS_NOMINAL_PASS`, while suprathreshold excursions trigger the emergency shutdown within the 2.5 ns budget. The worst sub-threshold detuning observed is $1.0000e-15$, and the sweep reports `STATUS_NOMINAL_PASS`.

D. SIMD-level phase rotation and zero-heap runtime

The HIL fatal-threshold path is implemented with the bare-metal AVX-512 instruction sequence `vmovaps`, `vcmpps`, `vmovmskps`, and `mov [mem], 0`, acting on a 64-byte-aligned stack-resident 16-lane `f32` buffer. The compare-extract-write chain takes roughly six clock cycles at 4.0 GHz, i.e. $\lesssim 1.5$ ns, so the full emergency shunt budget of 2.5 ns is never violated.

The U(1) phase-locked excitation $\psi_j \to e^{-i\theta_j}\psi_j$ is applied to an 8-component dark-ledger block. On `x86_64` the operation uses `vmovupd`, `vmulpd`, and `vfmadd231pd`; on `aarch64` it uses NEON `fmla` over 128-bit double lanes; otherwise a branchless scalar fallback is used. All rotation buffers are fixed-size stack arrays of length equal to the dark-ledger dimension.

Finally, the `rug` crate is compiled with custom GMP/MPFR memory functions installed via `mp_set_memory_functions`. Limb allocations are served from a pre-resident 16 MiB arena, removing variable-time `malloc`/`free` latencies from the 512-bit braiding and isometry verification loops.

E. Integrated 512-bit closure-chain audit

The completed levels of the $(26, 8, 312)$ branch are

$$I_\ell^\star = \frac{K}{2k_\ell} = \frac{312}{2\cdot 26} = 6.0000, \tag{29}$$

$$I_q^\star = \frac{K}{3k_q} = \frac{312}{3\cdot 8} = 13.0000. \tag{30}$$

The scalar framing defect is therefore

$$\Delta_{\mathrm{fr}} = I_q^\star - 2I_\ell^\star - 1 = 0.0, \tag{31}$$

so the closure chain (1) is satisfied: true. The global thermodynamic arrow is

$$\frac{dS_{\mathrm{total}}}{dt_{\mathrm{lc}}} = \Gamma_{\mathrm{de}} k_B \ln 2 + \frac{P_{\mathrm{cool}}}{T_c} = 0.0028\ \mathrm{K}^{-1}\mathrm{W}, \tag{32}$$

which is strictly positive (true). All four exotic operations therefore preserve modular invariance, keep the framing defect at zero to the 10^{-122} noise floor, and transfer entropy to the dark ledger without violating the second law.

Table II lists the HIL safety thresholds.

TABLE II. HIL safety thresholds.

Threshold	Value
Eigenvector rigidity	10^{-12}
Phase jitter	$5.0500e-05$ rad
Baseline temperature	0.0154 K
Emergency shunt latency	< 2.5 ns
Gate cycles at 72 GHz	180
Temperature rise	$3.7116e-43$ K

F. Acoustic impedance micro-engineering

The 142.08 MW field collapse is tamped by a single-crystal sapphire (Al_2O_3) waveguide with acoustic impedance $Z_{\mathrm{sapphire}} = 44.178$ MRayl. The peak acoustic pressure in the waveguide is

$$P_{\mathrm{peak}} = \sqrt{\frac{2P_{\mathrm{transient}} Z_{\mathrm{sapphire}}}{A_{\mathrm{waveguide}}}} \approx 12.6427\ \mathrm{GPa}, \tag{33}$$

using a waveguide area $A_{\mathrm{waveguide}} = \pi(5\,\mathrm{mm})^2$.

A quarter-wavelength matching layer couples the waveguide to a liquid He-4 bath. Its optimal impedance is the geometric mean

$$Z_m = \sqrt{Z_{\mathrm{sapphire}} Z_{\mathrm{He4}}} \approx 1.1512\ \mathrm{MRayl}, \tag{34}$$

which yields an effective liquid-helium impedance $Z_{\mathrm{He4}} \approx 0.03$ MRayl.

The acoustic pressure transmitted into the InP substrate at the waveguide/InP interface is

$$P_{\mathrm{InP}} = \frac{2Z_{\mathrm{InP}}}{Z_{\mathrm{sapphire}} + Z_{\mathrm{InP}}} P_{\mathrm{peak}} \approx 8.68 \text{ GPa}, \tag{35}$$

with $Z_{\mathrm{InP}} \approx 23.1$ MRayl. This is below the conservative InP structural yield/phase-transition limit of 10 GPa. The pressure entering the He-4 bath after the matching layer is ≈ 17 MPa, far below the cryostat design limit.

The simulator selects an alumina formulation based on frequency and thickness: AAO-Epoxy ($Z = 9.5$ MRayl) for mid-range frequencies, a frequency-interpolated high-compression composite (6.5–9.47 MRayl) for low frequencies, and colloidal nanocomposites for layers thinner than $10\,\mu$m.

VIII. ENGINEERING SYNTHESIS: RF PHASE CONTROL AND THERMAL FLUX MAPS

A. RF emitter-array phase-modulation table

The simulator exports a discrete phase-modulation table for an 8×8 InP/InGaAs SHBT emitter array. Each emitter is driven by a microwave command $e^{i\theta_{ij}}$ derived from the conformal dimension h_{ij} and the effective target velocity v_{eff}:

$$\theta_{ij} = 2\pi\, h_{ij}\, v_{\mathrm{eff}} \pmod{2\pi}. \tag{36}$$

The corresponding phase-shifter control voltage is linearly mapped between the gate/base turn-on voltage 3.8 V and the collector-drain supply 7.4 V. The exporter produces JSON/CSV tables with 64 entries per array; the live run reports phase-shifter voltages between 3.8000 V and 6.9500 V.

B. Thermal flux and cooling-power budget

The 14.2 μW refrigeration core is characterized by a de-rendering rate

$$\Gamma_{\mathrm{de}} = \frac{P_{\mathrm{cool}}}{k_B T_c \ln 2} = 9.6352e + 19 \text{ bit/s}, \tag{37}$$

giving a cooling power $P_{\mathrm{cool}} = 1.4200e - 05$ W. The average heat flux across the solid/He-4 interface is $q = 1.4200e + 07$ W/m^2.

For an un-engineered sapphire/He-4 boundary the acoustic-mismatch Kapitza resistance produces a temperature drop

$$\Delta T_{\mathrm{uneng}} = qR_K \approx 3.8900e + 14 \text{ K}, \tag{38}$$

while a quarter-wave Al_2O_3 matching layer reduces the effective mismatch factor and gives

$$\Delta T_{\mathrm{matched}} \approx 1.0657e + 13 \text{ K}. \tag{39}$$

The reduction $\Delta T_{\mathrm{matched}} < \Delta T_{\mathrm{uneng}}$ (true) justifies the Al_2O_3 acoustic-impedance engineering used in the HIL thermal model.

IX. CONCLUSION

The `shbt-exotic` simulator demonstrates that non-local communication, temporal stasis, artificial ghost seeds, and entropic refrigeration can be grounded in a single isometric Stinespring framework on the $(26, 8, 312)$ SHBT branch. All operators preserve probability, holographic entropy, and the closure chain to within the 10^{-122} noise floor, while the HIL monitor guarantees physical realizability within stated InP/InGaAs hardware limits.

CODE AVAILABILITY

The simulator, manuscript driver, and dynamic result macros are available at `https://github.com/sys1own/shbt-exotic`. Related SHBT engines are at `https://github.com/sys1own/shbt-precision`, `https://github.com/sys1own/shbt-warp`, and `https://github.com/sys1own/shbt-recon`.